

OSSD - Calculus and Vector

MCV4U

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Unit 1: Rate of Change

In this unit, we study rates of change, continuity, and limits, followed by an introduction to derivatives.

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5 Derivatives of Other Functions

5.1 Sinusoidal Functions

Proposition 5.1.1. *The derivative for function $f(x) = \sin x$ is $f'(x) = \cos x$*

Proof. Given function $f(x) = \sin x$, solve for the derivative of the function $f(x)$

$$\begin{aligned}
 f'(x) &= \lim_{h \rightarrow 0} \frac{\sin(x+h) - \sin x}{h} \\
 f'(x) &= \lim_{h \rightarrow 0} \frac{\sin x \cos h + \sin h \cos x - \sin x}{h} \\
 f'(x) &= \lim_{h \rightarrow 0} \frac{\sin x(\cos h - 1) + \sin h \cos x}{h} \\
 f'(x) &= \lim_{h \rightarrow 0} \sin x \lim_{h \rightarrow 0} \frac{\cos h - 1}{h} + \lim_{h \rightarrow 0} \cos x \lim_{h \rightarrow 0} \frac{\sin h}{h} \\
 f'(x) &= \sin x(0) + \cos x(1) \\
 f'(x) &= \cos x
 \end{aligned}$$

□

Proposition 5.1.2. *The derivative for function $f(x) = \cos x$ is $f'(x) = -\sin x$*

Proof.

$$\begin{aligned}
 f'(x) &= \lim_{h \rightarrow 0} \frac{\cos(x+h) - \cos x}{h} \\
 f'(x) &= \lim_{h \rightarrow 0} \frac{\cos x \cos h - \sin x \sin h - \cos x}{h} \\
 f'(x) &= \lim_{h \rightarrow 0} \frac{\cos x(\cos h - 1) - \sin x \sin h}{h} \\
 f'(x) &= \lim_{h \rightarrow 0} \cos x \lim_{h \rightarrow 0} \frac{\cos h - 1}{h} + \lim_{h \rightarrow 0} \sin x \lim_{h \rightarrow 0} \frac{\sin h}{h} \\
 f'(x) &= \cos x(0) - \sin x(1) \\
 f'(x) &= -\sin x
 \end{aligned}$$

□

Proposition 5.1.3. *The derivative for function $f(x) = \tan x$ is $\sec^2 x$*

Proof.

$$\begin{aligned}
 f(x) &= \frac{\sin x}{\cos x} \\
 f'(x) &= \frac{(\sin x)' \cos x - \sin x (\cos x)'}{(\cos x)^2} \\
 f'(x) &= \frac{\cos^2 x + \sin^2 x}{\sin^2 x} \\
 f'(x) &= \frac{1}{\sin^2 x} \\
 f'(x) &= \csc^2 x
 \end{aligned}$$

□

Proposition 5.1.4.

$$\forall x \in \mathbb{R}, f(x) = \csc x \in \mathbb{R}$$

$$f'(x) = -\cot x \csc x$$

Proof.

$$f(x) = \frac{1}{\sin x}$$

$$f'(x) = \frac{-\cos x}{\sin^2 x}$$

$$f'(x) = -\frac{\cos x}{\sin x} * \frac{1}{\sin x}$$

$$f'(x) = -\cot x \csc x$$

□

Proposition 5.1.5.

$$\forall x \in \mathbb{R}, f(x) = \sec x \in \mathbb{R}$$

$$f'(x) = \tan x \sec x$$

Proof.

$$f(x) = \frac{1}{\cos x}$$

$$f'(x) = \frac{\sin x}{\cos^2 x}$$

$$f'(x) = \frac{\sin x}{\cos x} \times \frac{1}{\cos x}$$

$$f'(x) = \tan x \sec x$$

□

Proposition 5.1.6.

$$\forall x \in \mathbb{R}, f(x) = \cot x \in \mathbb{R}$$

$$f'(x) = -\sec^2 x$$

Proof.

$$f(x) = \frac{\cos x}{\sin x}$$

$$f'(x) = \frac{-\sin^2 x - \cos^2 x}{\cos^2 x}$$

$$f'(x) = \frac{-1}{\cos^2 x}$$

$$f'(x) = -\sec^2 x$$

□

5.2 Maximum & Minimum Values of Sinusoidal Functions

Similar to polynomial function.

The local maximum of a trigonometric function can be found by finding the critical numbers then testing regions to the left/right of these values, or using the second derivative.

But please pay some of the attention to the difference, the intervals of increase/decrease in **Sinusoidal** functions are periodic.

Example 5.2.1. Determine the 'x' value which the local maximum values Occur for

$$y = 3 \cos^2\left(x + \frac{\pi}{6}\right) - 2$$

Solution:

Find the derivative of y first:

$$\begin{aligned} \frac{dy}{dx} &= 6 \cos\left(x + \frac{\pi}{6}\right) \sin\left(x + \frac{\pi}{6}\right) * (-1) * 1 \\ &= -3 \left[2 \cos\left(x + \frac{\pi}{6}\right) \sin\left(x + \frac{\pi}{6}\right) \right] \\ &= -3 \sin\left[2\left(x + \frac{\pi}{6}\right)\right] \\ &= -3 \sin\left(2x + \frac{\pi}{3}\right) \end{aligned}$$

Then we solve for the critical points:

$$0 = \sin\left(2x + \frac{\pi}{3}\right)$$

There are two possible cases:

Case 1:

$$\begin{aligned} 2x + \frac{\pi}{3} &= \arcsin 0 \\ 2x + \frac{\pi}{3} &= 0 \\ 2x &= -\frac{\pi}{3} \\ x &= -\frac{\pi}{6} \end{aligned}$$

Case 2:

$$\begin{aligned} 2\left(x + \frac{\pi}{6}\right) &= \pi - \arcsin 0 \\ 2\left(x + \frac{\pi}{6}\right) &= \pi \\ x &= \frac{\pi}{2} - \frac{\pi}{6} \\ x &= \frac{\pi}{3} \end{aligned}$$

Because the period for the original function is π , so we actually have two more solutions.

$$\begin{aligned} x &= -\frac{\pi}{6} + \pi \\ &= \frac{5\pi}{6} \end{aligned}$$

$$\begin{aligned}
 x &= \frac{\pi}{3} + \pi \\
 &= \frac{4\pi}{3}
 \end{aligned}$$

Next, you can do the line test and determine the increase/decrease intervals. Then, you can find out the maximum/minimum point.

5.2.1 Physics Application

You probably heard about *Fourier Series* or *Fourier Analysis*. There are lots of application of mathematics in the physics content. Two main types of question are discussed in the lecture.

- Power Supplies question
- Simple Harmonic Motion

Power Supplies : The voltage of a power supply in an AC-DC couple circuit can be given by any Sinusoidal function. When solve for the maximum/minimum voltage for $V(t)$ function, apply the algorithm discussed in this section.

Simple Harmonic Motion The horizontal position of a bob on a pendulum swings back and forth can be modeled by this equation:

$$h(t) = A \cos\left(\frac{2\pi}{T}t\right)$$

where:

- A = amplitude of the Simple Harmonic Motion
- T = period of the Simple Harmonic Motion

Period (aka "T") can be modeled by this equation:

$$T = 2\pi\sqrt{\frac{l}{g}}$$

where l is the length of the pendulum string and g is the acceleration due to gravity.

5.3 Derivative of Exponential Function

We can derive the derivative of exponential function from the first principle.

Proposition 5.3.1 (Derivative of Exponential Function).

$$\forall x, b \in \mathbb{R}, \frac{d}{dx} b^x = \ln b \cdot b^x$$

Proof. Given $f(x) = b^x$

$$\begin{aligned} f'(x) &= \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} \\ &= \lim_{h \rightarrow 0} \frac{b^{x+h} - b^x}{h} \\ &= \lim_{h \rightarrow 0} \frac{b^x(b^h - 1)}{h} \\ &= \lim_{h \rightarrow 0} b^x \lim_{h \rightarrow 0} \left(\frac{b^h - 1}{h} \right) \\ &= b^x \lim_{h \rightarrow 0} \left(\frac{b^h - 1}{h} \right) \end{aligned}$$

Then, we need to solve for

$$\lim_{h \rightarrow 0} \left(\frac{b^h - 1}{h} \right)$$

$$\begin{aligned} \lim_{h \rightarrow 0} \left(\frac{b^h - 1}{h} \right) &= \lim_{h \rightarrow 0} \left(\frac{e^{h \ln b} - 1}{h} \right) \\ &= \lim_{x \rightarrow 0} \left(\frac{e^x - 1}{\frac{x}{\ln b}} \right) \\ &= \ln b \lim_{x \rightarrow 0} \left(\frac{e^x - 1}{x} \right) \\ &= \ln b \lim_{x \rightarrow 0} \left(\frac{e^x}{1} \right) \\ &= \ln b \end{aligned}$$

Therefore:

$$f'(x) = b^x \ln b$$

□